

Scientific Justification *Limited to 1 page of text plus 2 pages for figures and references.*

Challenges to standard cosmology: Over the last three decades, Λ CDM has emerged as our standard cosmological paradigm. While Λ , the cosmological constant and cold dark matter together comprise $\sim 95\%$ of the universe, their physical origin is poorly understood. The *Hubble tension* - a $> 5\sigma$ discrepancy between the early universe inference, assuming the standard model and the local Cepheid distance ladder measurement has emerged as a serious conundrum [1,2]. Moreover, recent hints that dark energy maybe evolving with cosmic time is contrary to the cosmological constant scenario [3]. However, there is a key open question as to whether these results are a sign of new cosmological physics or unknown systematics. Type Ia supernovae (SNe Ia) are central to both measurements and hence, play a starring role in modern cosmology. Strong gravitationally lensed SNe Ia (gLSNe Ia) are, excellent laboratories to address both open questions re: systematics in dark energy and H_0 measurements. The “gravitational telescope” effect of lensing magnification allows us to study high- z SNe Ia in much greater detail than unlensed SNe at the same redshift. Comparing properties of high- z SNe to local counterparts is a vital test for whether there is a change in intrinsic properties of SNe Ia with redshift or “cosmological drift”, which can mimic a signal of dark energy. gLSNe measure the Hubble Constant via the delay between the arrival time of the multiple images - making them independent of the current leading probes of H_0 [4]. With a modest amount of Gemini time, we aim to leverage both these advantages of gLSNe Ia to understand SN Ia cosmic evolution and measure H_0 .

As gLSNe are rare, finding them requires wide-field imaging surveys. The Zwicky Transient Facility (ZTF) surveys entire Northern sky with a 2-day cadence down to ~ 20.5 mag in *gr* filters - along with *i*-band from partnership time - ideal for rare transient discovery, as demonstrated by the discovery of three lensed SNe [5-7]. **The discovery of SN2026ngr - a lensed SN Ia at $z = 0.95$ [8], a lookback time of $\sim 8\text{Gyr}$ - presents a new window for the first UV spectral time series of a $z \sim 1$ SN Ia.** As the images are separated by $\sim 1.2''$, Gemini imaging will be important to measure time-delays.

Restframe UV spectroscopy: Dark energy systematics test SNe Ia measure dark energy properties, e.g. its present-day equation of state (w_0) and the time-dependence (w_a) via a *relative* distance measurement between high- z ($z \sim 1$) and low- z ($z \sim 0.1$) objects [9-11]. If there are intrinsic differences between the populations of SNe Ia at low- and high- z it can appear as a signal for evolving dark energy, when the true cosmology is not one where dark energy evolves [12]. Testing for evolution requires comparing features of SNe Ia that are distance independent between low- and high- z populations, e.g., spectral lines. As distances are typically measured in the observer-frame optical (e.g. *griz* filters), this corresponds to the rest-frame *UV*. Therefore, it is crucial to compare SN Ia spectra at high- z in the UV with their low- z counterparts. The ultraviolet spectra of SNe Ia are dominated by Fe-group element lines, and are a tracer of progenitor metallicity. Such tests have typically only been carried out at near peak phases and / or $z \sim 0.5$ with possible signs hinting at differences [13-15]. However, since the SNIa Hubble diagram extends to $z \sim 1$, SN2026ngr is a unique opportunity to compare the SNe Ia at the highest redshift to $z \lesssim 0.1$ SNe Ia. In the absence of lensing magnifications, these SNe would be too faint to get spectra even from 8m class telescopes.

Time-delays for H_0 : Measuring H_0 from gLSNe requires two main ingredients, a high-resolution image of the lensing system and a time-delay between multiple images. As SNe Ia are standardisable candles, we can observationally infer their peak magnitude, and hence, lensing magnification. This breaks the mass-sheet degeneracy which is an important systematic that we need to overcome in lens modelling [15]. Moreover, the well understood family of lightcurves of SNe Ia help in accurately inferring time-delays with only a handful of epochs [16], which this proposal will provide. **SN2026ngr provides the first opportunity to measure H_0 precisely with a galaxy-scale lensed SN Ia, as previous galaxy scaled gLSNe Ia had extremely short time-delays ([5,17]).** Cadenced observations with rest-frame ~ 1 week in *riz* filters will allow us to measure the decline post-peak in the optical, including the shoulder in the *V*-band. This feature is critical to measure the arrival time difference between the multiple images.

References: [1] Planck Collaboration, 2020, A&A, 641, A6; [2] Riess et al., 2022, ApJ, 934, 7; [3] DESI Collaboration, Abdul Karim et al., 2025, PRD, 112, 8; [4] Refsdal, 1964, MNRAS, 128, 37; [5] Goobar et al., 2023, NatAst, 7, 1098; [6] Johansson et al. 2025, ApJ, 995, 17; [7] Lemon et al. 2026, ApJ accepted; [8] Johansson & Goobar, 2026, AstroNote, 2026TNSAN.156....1J; [9] Brout et al. 2022, ApJ, 938, 110; [10] Rubin et al. 2025, ApJ, 986, 231; [11] DES Collaboration: Abbott et al., 2025, ApJ, 973, 14; [12] Dhawan et al. 2025, MNRAS, 540, 1626; [13] Foley et al. 2008, ApJ, 684, 1; [14] Maguire et al. 2012, MNRAS, 426, 3; [15] Ellis et al. 2008, ApJ, 674, 51; [16] Birrer et al. 2022, ApJ, 924, 2; [17] Hayes et al. 2026, MNRAS, 546, 113; [18] Goobar et al. 2017, Science, 356, 291; [19] Birrer et al. 2020, A&A 643, A165; [20] Pierel et al. 2022, ApJ, 939, 11;

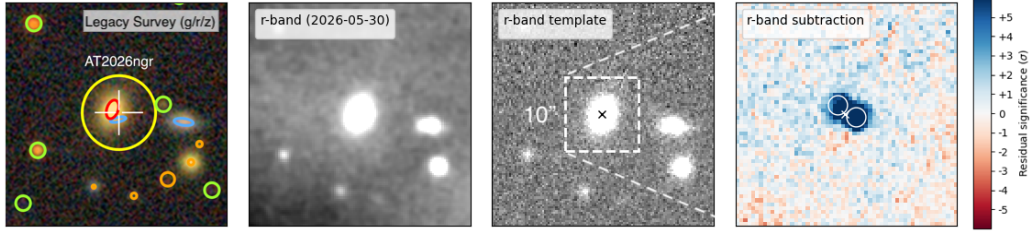


Figure 1: r -band image from the Gran Telescopio Canarias OSIRIS obtained on 2026-06-01. The scaled/shifted science, reference and difference images are shown. From the zoomed-in difference image we can see two blended images suggesting that the system would be lensed. The SN has r, i, z magnitudes of 20.8, 19.3, 19.0 mag, respectively, with a typical SN Ia decline of $\sim 1 - 1.5$ mag in 15 restframe days. Hence for spectroscopy and image time series we need an 8m class telescope.

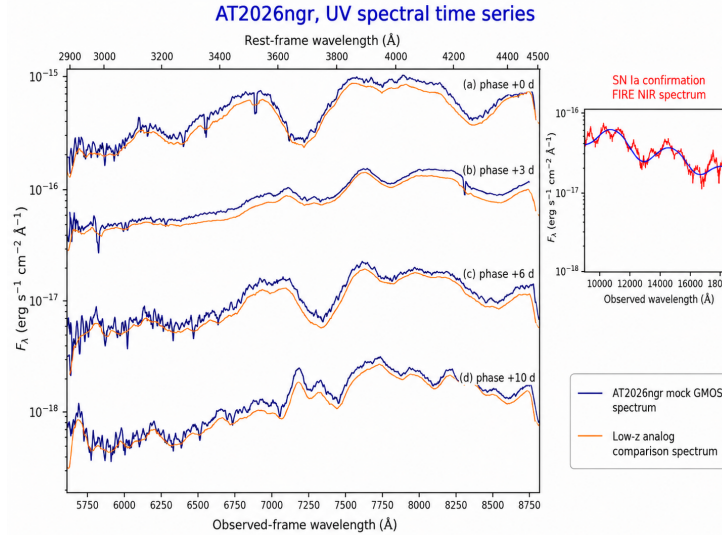


Figure 2: Near-UV spectra of the local sample (orange) from [13] at various phases post maximum light. Mock spectra of SN2026ngr with the expected SNR from GMOS. The inset shows the classification spectrum. Any differences in the features, e.g. velocities and their evolution (see Fig. 3) will suggest cosmic evolution of SN Ia properties.

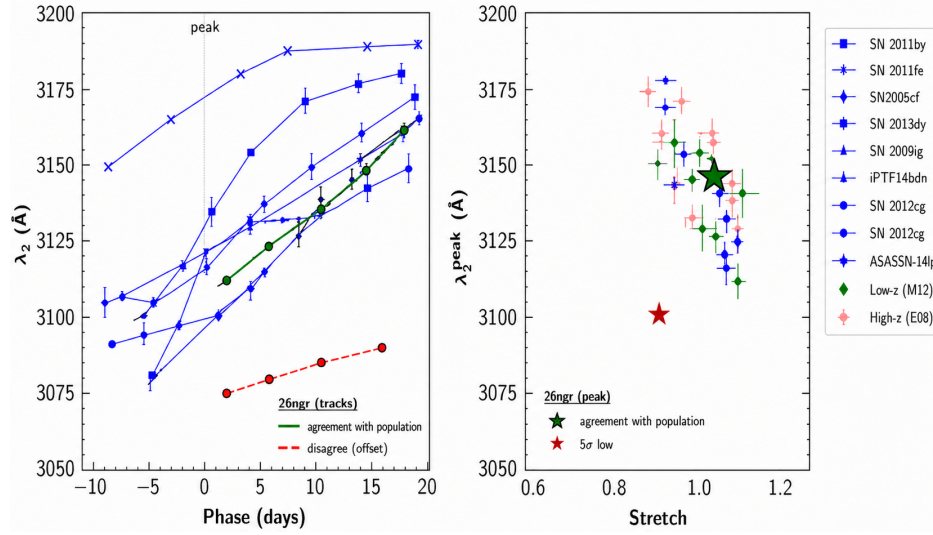


Figure 3: (Left) The evolution of the λ_1 feature in the UV spectra versus phase. The green shows the case of SN2026ngr being consistent with the low- z population and the red shows the case if its discrepant. (Right) The position of the λ_2 feature at peak brightness versus lightcurve stretch (SNe get brighter towards the right) for the low- and intermediate- z sample. The green point shows the case of no cosmological evolution and the orange point shows strong disagreement from the trend.

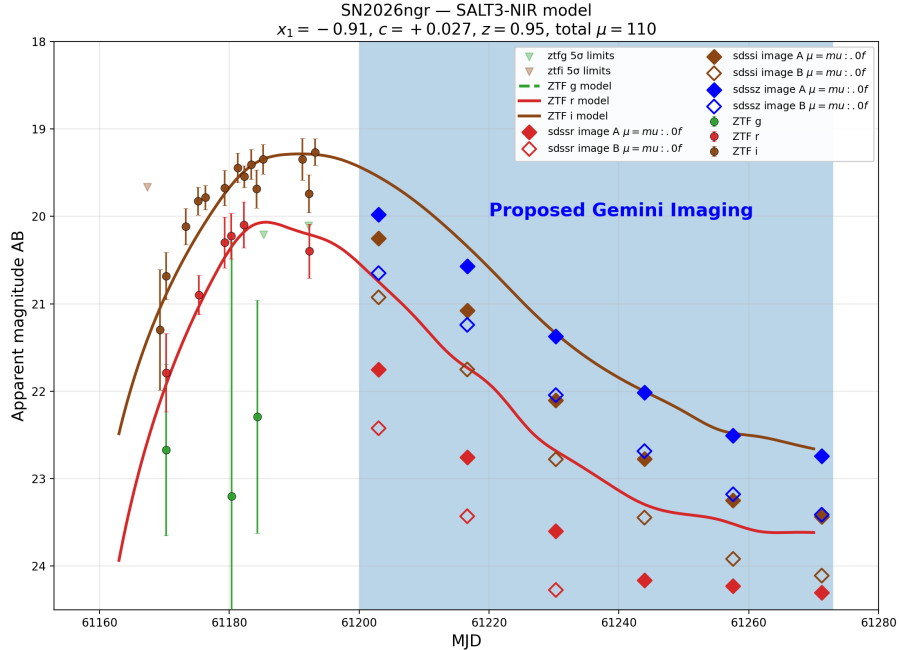


Figure 4: Current lightcurve of SN2026ngr from unresolved observations showing the SN is rising to peak in the i -band. The simulated lightcurves show riz from GMOS with a week's cadence in the observer frame. The resolved data are crucial for accurate time-delays using *Glimpse* [16]. The model in the i -band is extrapolated using the SALT3-NIR fit [20]

Experimental Design *Limited to 1 page of text*

SN2026ngr is an ideal system for testing cosmic evolution of SN Ia properties and measuring H_0 . It is at $z = 0.95$ and has a high magnification of $\mu \sim 100$, derived from comparing the unresolved peak magnitude to that of an unlensed SN Ia at the same redshift. To keep a future H_0 analysis to be blind, we intentionally avoid computing a more precise model predicted time-delay (e.g. using the full imaging information available). We describe our experimental design for optical spectroscopy and photometry of SN2026ngr. We propose to get the first time-series of restframe UV spectra of a $z \sim 1$ SN Ia. We request a weekly cadence in the observer frame to trace the spectral features in the UV. **While this is a single object, the results will pave the way for building the sample with the Vera C. Rubin Observatory and the Roman Space Telescope. Moreover, we expect that since these surveys have deeper limiting magnitudes the magnification distribution will peak lower than the value for SN2026ngr, therefore, this SN is a unique laboratory to get high-fidelity spectroscopy and imaging.**

UV Spectroscopic Comparison We propose to obtain one GMOS-S spectrum every week in the observer frame. This corresponds to $\sim 3 - 4$ d in the restframe of the SN. Getting a total of 4 spectra allows us to test spectral evolution thoroughly between $\sim +5$ d and $\sim +15$ d, phases for which low- z SNe have been observed. We will compare the low- z spectra to the time series obtained here by creating a mean spectrum of the low- z sample, as done in [16,17,18]. There is a large dataset of UV spectra at different phases in the nearby universe. Mean spectra will be created in phase bins of 3d (e.g. like the SNe shown in fig. 2). Based on the dispersion around the mean spectra, we can find deviations or lack thereof from the observations of SN2026ngr. Another test of spectral evolution is via line positions, velocities and pseudo equivalent widths of specific features. In the UV region, we will measure the position of the two peaks (dubbed λ_1 and λ_2 in [14]) in the region from 2900 - 3300 Å, which has been shown to correlate with lightcurve shape and the Ca H&K velocity. Comparing the inferred values for SN2026ngr with the low- z λ_2 -stretch will allow us to test whether SN2026ngr is discrepant with the local population of SNe Ia.

Seeing pronounced differences in the mean spectra or the position of the peaks of the Fe features in the UV could suggest intrinsic differences between SN2026ngr, whereas similarity between the spectra would suggest that cosmic evolution is minimal. **These spectra - first of their kind - will set the framework for building a high- z spectral time series with a sample of lensed SNe Ia, e.g. from the Vera C. Rubin Observatory.**

Photometric Time-Delays High-resolution, cadenced photometry in multiple filters is necessary to infer an accurate time-delay from lightcurves [14]. The *griz* data from GMOS will be fitted with `Glimpse`¹ in combination with unresolved photometry. For the proposed six epochs of resolved imaging, we find that we can infer time-delays with an accuracy ~ 0.7 d. Assuming a true time-delay of ~ 10 d that allows us a measurement at 7% level which is comparable to the error budget on H_0 from the most accurate lensed SN study to date [14]. The `glimpse` fit uncertainties include marginalising microlensing and dust extinction effects.

Final H_0 inference: Cadenced photometry of the individual images is the only missing piece for cosmography with SN2026ngr. Single epoch high-resolution imaging will be obtained for this SN from already obtained telescope time to model the lensing system. The lens redshift is reported by the Dark Energy Spectroscopic Instrument [10] and the source redshift is 0.95. The time-delays - from the combination of photometry obtained from this proposal will then be used with the lens model to infer H_0 with a $\sim 8\%$ error using cosmographic inference tools, e.g. `hierarc` [20].

¹<https://github.com/erinhay/GausSN/tree/glimpse>

Technical Description *Limited to 1 page of text*

We request Gemini South observations as this is an equatorial target and Gemini North is currently not operational.

We request 6 epochs of r,i,z photometry with GMOS-S with a one week cadence in the restframe. Using the Gemini ITC we take the expected SNR for the faintest image which has an $m_R = 20.8$ mag at present. To get an S/N of 20 for the faintest image, we request 2 X 60s for the g,z -band 2 X 45s on source in each of the r,i filters. We assume 70% image quality, median water vapor, 50% cloud cover and 50% sky brightness. Including preset, telescope offset, readout time and DHS write we request a total of **30 minutes per epoch for a total of 3h of photometry**. We assume the underlying source to be an SNIa at $z = 0.95$.

For spectroscopy, since the SN has broad features, we request We use the GMOS R400 G5305 grating, which provides $R = 1500$ -1900 for a 1 slit with a central wavelength of 700nm. We assume image A to be $r \sim 20.8$ mag. For the measurement error to be subdominant in the velocity gradient inference, we require $\sigma(v) < 300 \text{ km s}^{-1}$ in the *rest frame*. To get sufficient S/N (≥ 10 per spectral pixel at $\lambda \sim 7000 \text{ \AA}$) for an accurate measurement of the age, we request **2 X 900s** on source for 70% image quality, median water vapor, 50% cloud cover and 50% sky brightness. Since it is a conservative upper limit on the S/N, we can take observations in 85% IQ or gray time. Including setup, offset, readout and DHS write, this corresponds to **45 minutes per epoch for a total of 3h of spectroscopy for four epochs**.

Therefore, we request a total of **6h** of observing time with GMOS.

DDT Justification *Limited to 1/2 page of text*

As SN2026ngr is at peak brightness, these observations are extremely time critical. SN2026ngr was discovered after the last queue cycle for which we could apply and get data now. There no existing program for spectral time series of lensed transients and fast turnaround would take > 1 month for approval, by when the transient would've faded below detection, also true for the next queue cycle. SN2026ngr is the first system for an unprecedented test of cosmological evolution of SNe Ia and with an appreciable time-delay for the possibility of measuring H_0 with a galaxy-scale lensed SN Ia. DD time is critical for such high-impact observations. **Due to the fading rate of SNe Ia and the need for accurate time-delays this is a time-critical proposal.**

Band 3 Plan *Limited to 1/2 page of text*

N/A

Classical Backup Program *Limited to 1/2 page of text*

N/A

Justify Target Duplications

This is a new discovery with no Gemini data in the archive.

ITC Examples

ITC outputs for spectroscopy using the R400 grating and griz imaging. We estimate the total time request per epoch (see attached ITC outputs) for photometry by removing the common overheads per epoch i.e. of telescope setup and offset.

Gemini Integration Time Calculator

GMOS-S - 2026B.1.1.1

[Click here for help with the results page.](#)

Read noise: 3.98 e-/pix

Dark current: 8.3E-4 e-/s/pix

software aperture extent along slit = 1.12 arcsec

fraction of source flux in aperture = 0.76

derived image size(FWHM) for a point source = 0.80 arcsec

Sky subtraction aperture = 5.0 times the software aperture.

Requested total integration time = 1500.00 secs, of which 1500.00 secs is on source.

The peak pixel signal + background is 1465 e- (813 ADU). This is 1% of the saturation limit of 117963 e-.

Observation Overheads

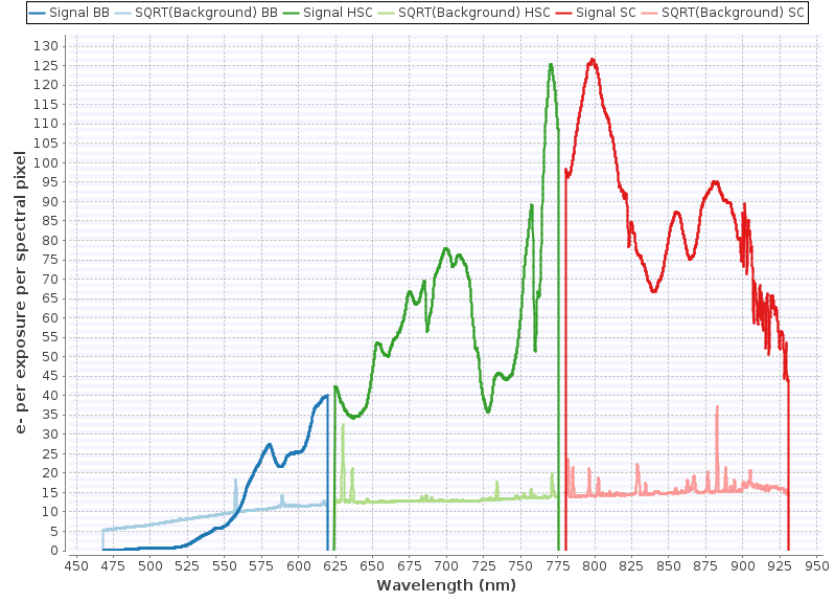
Setup	960.0 s	
Telescope offset	1 x 7.0 s	assuming ABBA dithering pattern
Exposure	3 x 500.0 s	
Readout	3 x 50.3 s	
DHS Write	3 x 10.0 s	
Program time	44 mins 8 secs	

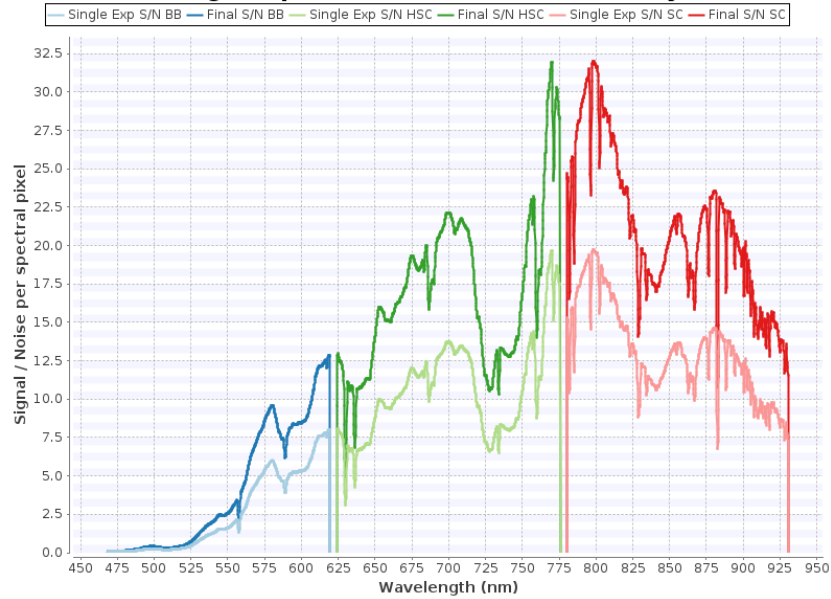
[Click here for ASCII signal spectrum.](#)

[Click here for ASCII background spectrum.](#)

[Click here for Single Exposure S/N ASCII data.](#)

[Click here for Final S/N ASCII data.](#)

Signal and SQRT(Background) in one pixel

Intermediate Single-Exposure S/N and Final S/N with Sky Subtraction**Output:**

- Spectra autoscaled.

Input Parameters:

Instrument: GMOS-S

Source spatial profile, brightness, and spectral distribution:

The $z = 0.95000$ point source is a 19.6 Vega user defined SED in the i band with the name: sn2011fe_exptime.dat

Instrument configuration:**Optical Components:**

- Fixed Optics
 - Grating Optics: R400_G5325
 - Detector - Hamamatsu array
- Amp gain: Low, Amp read mode: Slow
- Focal Plane Mask: Longslit 1.00 arcsec

Region of Interest: Full Frame Readout

Central Wavelength: 700.0 nm

Binning: 2×1

Pixel Size in Spatial Direction: 0.080778arcsec

Pixel Size in Spectral Direction: 0.148nm

Telescope configuration:

- silver mirror coating.
- side looking port.

Gemini Integration Time Calculator

GMOS-S - 2026B.1.1.1

[Click here for help with the results page.](#)

software aperture diameter = 0.94 arcsec
fraction of source flux in aperture = 0.62
enclosed pixels = 107.26

derived image size for Gaussian source = 0.80

Sky subtraction aperture = 5.0 times the software aperture.
Read noise: 3.98 e-/pix
Dark current: 8.3E-4 e-/s/pix

S/N for BB:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 55.14
Background noise = 428.70
Dark current noise = 2.31
Readout noise = 41.22

Total noise per exposure = 434.20
Total signal per exposure = 3040.75

Intermediate S/N for one exposure = 7.00

S/N for the whole observation = 9.05 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 1751 e- (972 ADU). This is 1% of the saturation limit of 117963 e-.

S/N for HSC:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 58.11
Background noise = 445.46
Dark current noise = 2.31
Readout noise = 41.22

Total noise per exposure = 451.13
Total signal per exposure = 3376.85

Intermediate S/N for one exposure = 7.49

S/N for the whole observation = 9.68 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 1892 e- (1051 ADU). This is 2% of the saturation limit of 117963 e-.

S/N for SC:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 53.38

Background noise = 400.54

Dark current noise = 2.31

Readout noise = 41.22

Total noise per exposure = 406.18

Total signal per exposure = 2848.96

Intermediate S/N for one exposure = 7.01

S/N for the whole observation = 9.07 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 1531 e- (850 ADU). This is 1% of the saturation limit of 117963 e-.

Observation Overheads

Setup	360.0 s	
Telescope offset	1 x 7.0 s	assuming ABAB dithering pattern
Exposure	2 x 60.0 s	
Readout	2 x 82.5 s	
DHS Write	2 x 10.0 s	
Program time	11 mins 12 secs	

Output:

- Spectra autoscaled.

Input Parameters:

Instrument: GMOS-S

Source spatial profile, brightness, and spectral distribution:

The z = 0.95000 point source is a 19.6 Vega user defined SED in the i band with the name: sn2011fe_exptime.dat

Instrument configuration:

Optical Components:

- Filter: g_G0325
- Fixed Optics
- Detector - Hamamatsu array

Amp gain: Low, Amp read mode: Slow

Region of Interest: Full Frame Readout

Binning: 1 x 1

Pixel Size in Spatial Direction: 0.080778arcsec

Telescope configuration:

- silver mirror coating.
- side looking port.

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Gemini Integration Time Calculator

- wavefront sensor: oiwfs

Observing Conditions:

- Airmass: 1.50
- Image Quality: 70% ($\leq 0.63''$ at zenith, $\leq 0.80''$ on-source, **exact condition specified**)
- Cloud cover: 70%
- Water Vapor: 100%
- Sky Background: 80%

Likelihood of execution: 39%

Calculation and analysis methods:

- Mode: imaging
- Calculation of S/N ratio with 2 exposures of 60.00 secs, and 100.00% of them on source.
- Analysis performed for aperture that gives 'optimum' S/N and a sky aperture that is 5.00 times the target aperture.

Gemini Integration Time Calculator

GMOS-S - 2026B.1.1.1

[Click here for help with the results page.](#)

software aperture diameter = 0.94 arcsec
fraction of source flux in aperture = 0.62
enclosed pixels = 107.26

derived image size for Gaussian source = 0.80

Sky subtraction aperture = 5.0 times the software aperture.
Read noise: 3.98 e-/pix
Dark current: 8.3E-4 e-/s/pix

S/N for BB:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 233.20
Background noise = 478.04
Dark current noise = 2.31
Readout noise = 41.22

Total noise per exposure = 533.48
Total signal per exposure = 54380.52

Intermediate S/N for one exposure = 101.93

S/N for the whole observation = 133.74 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 2803 e- (1557 ADU). This is 2% of the saturation limit of 117963 e-.

S/N for HSC:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 236.58
Background noise = 487.43
Dark current noise = 2.31
Readout noise = 41.22

Total noise per exposure = 543.38
Total signal per exposure = 55971.15

Intermediate S/N for one exposure = 103.01

S/N for the whole observation = 135.13 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 2907 e- (1615 ADU). This is 2% of the saturation limit of 117963 e-.

S/N for SC:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 231.18

Background noise = 474.25

Dark current noise = 2.31

Readout noise = 41.22

Total noise per exposure = 529.21

Total signal per exposure = 53444.90

Intermediate S/N for one exposure = 100.99

S/N for the whole observation = 132.50 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 2758 e- (1532 ADU). This is 2% of the saturation limit of 117963 e-.

Observation Overheads

Setup	360.0 s	
Telescope offset	1 x 7.0 s	assuming ABAB dithering pattern
Exposure	2 x 60.0 s	
Readout	2 x 82.5 s	
DHS Write	2 x 10.0 s	
Program time	11 mins 12 secs	

Output:

- Spectra autoscaled.

Input Parameters:

Instrument: GMOS-S

Source spatial profile, brightness, and spectral distribution:

The z = 0.95000 point source is a 19.6 Vega user defined SED in the i band with the name: sn2011fe_exptime.dat

Instrument configuration:

Optical Components:

- Filter: r_G0326
- Fixed Optics
- Detector - Hamamatsu array

Amp gain: Low, Amp read mode: Slow

Region of Interest: Full Frame Readout

Binning: 1×1

Pixel Size in Spatial Direction: 0.080778arcsec

Telescope configuration:

- silver mirror coating.
- side looking port.

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Gemini Integration Time Calculator

- wavefront sensor: oiwfs

Observing Conditions:

- Airmass: 1.50
- Image Quality: 70% ($\leq 0.63''$ at zenith, $\leq 0.80''$ on-source, **exact condition specified**)
- Cloud cover: 70%
- Water Vapor: 100%
- Sky Background: 80%

Likelihood of execution: 39%

Calculation and analysis methods:

- Mode: imaging
- Calculation of S/N ratio with 2 exposures of 60.00 secs, and 100.00% of them on source.
- Analysis performed for aperture that gives 'optimum' S/N and a sky aperture that is 5.00 times the target aperture.

Gemini Integration Time Calculator

GMOS-S - 2026B.1.1.1

[Click here for help with the results page.](#)

software aperture diameter = 0.94 arcsec
fraction of source flux in aperture = 0.62
enclosed pixels = 107.26

derived image size for Gaussian source = 0.80

Sky subtraction aperture = 5.0 times the software aperture.
Read noise: 3.98 e-/pix
Dark current: 8.3E-4 e-/s/pix

S/N for BB:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 320.05
Background noise = 521.32
Dark current noise = 2.31
Readout noise = 41.22

Total noise per exposure = 613.11
Total signal per exposure = 102431.38

Intermediate S/N for one exposure = 167.07

S/N for the whole observation = 220.75 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 3801 e- (2111 ADU). This is 3% of the saturation limit of 117963 e-.

S/N for HSC:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 309.52
Background noise = 504.48
Dark current noise = 2.31
Readout noise = 41.22

Total noise per exposure = 593.30
Total signal per exposure = 95803.60

Intermediate S/N for one exposure = 161.48

S/N for the whole observation = 213.36 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 3558 e- (1976 ADU). This is 3% of the saturation limit of 117963 e-.

S/N for SC:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 311.98
 Background noise = 508.26
 Dark current noise = 2.31
 Readout noise = 41.22

Total noise per exposure = 597.80
 Total signal per exposure = 97328.82

Intermediate S/N for one exposure = 162.81

S/N for the whole observation = 215.13 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 3612 e- (2006 ADU). This is 3% of the saturation limit of 117963 e-.

Observation Overheads

Setup	360.0 s	
Telescope offset	1 x 7.0 s	assuming ABAB dithering pattern
Exposure	2 x 60.0 s	
Readout	2 x 82.5 s	
DHS Write	2 x 10.0 s	
Program time	11 mins 12 secs	

Output:

- Spectra autoscaled.

Input Parameters:

Instrument: GMOS-S

Source spatial profile, brightness, and spectral distribution:

The z = 0.95000 point source is a 19.6 Vega user defined SED in the i band with the name: sn2011fe_exptime.dat

Instrument configuration:

Optical Components:

- Filter: i_G0327
 - Fixed Optics
 - Detector - Hamamatsu array
- Amp gain: Low, Amp read mode: Slow

Region of Interest: Full Frame Readout

Binning: 1×1

Pixel Size in Spatial Direction: 0.080778arcsec

Telescope configuration:

- silver mirror coating.
- side looking port.

02/06/2026, 19:20

Gemini Integration Time Calculator

- wavefront sensor: oiwfs

Observing Conditions:

- Airmass: 1.50
- Image Quality: 70% ($\leq 0.63''$ at zenith, $\leq 0.80''$ on-source, **exact condition specified**)
- Cloud cover: 70%
- Water Vapor: 100%
- Sky Background: 80%

Likelihood of execution: 39%

Calculation and analysis methods:

- Mode: imaging
- Calculation of S/N ratio with 2 exposures of 60.00 secs, and 100.00% of them on source.
- Analysis performed for aperture that gives 'optimum' S/N and a sky aperture that is 5.00 times the target aperture.

Gemini Integration Time Calculator

GMOS-S - 2026B.1.1.1

[Click here for help with the results page.](#)

software aperture diameter = 0.94 arcsec
fraction of source flux in aperture = 0.62
enclosed pixels = 107.26

derived image size for Gaussian source = 0.80

Sky subtraction aperture = 5.0 times the software aperture.
Read noise: 3.98 e-/pix
Dark current: 8.3E-4 e-/s/pix

S/N for BB:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 298.26
Background noise = 749.96
Dark current noise = 2.31
Readout noise = 41.22

Total noise per exposure = 808.15
Total signal per exposure = 88959.31

Intermediate S/N for one exposure = 110.08

S/N for the whole observation = 143.75 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 6344 e- (3524 ADU). This is 5% of the saturation limit of 117963 e-.

S/N for HSC:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 282.92
Background noise = 710.74
Dark current noise = 2.31
Readout noise = 41.22

Total noise per exposure = 766.10
Total signal per exposure = 80044.28

Intermediate S/N for one exposure = 104.48

S/N for the whole observation = 136.45 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 5700 e- (3166 ADU). This is 5% of the saturation limit of 117963 e-.

S/N for SC:

Contributions to total noise (e-) in aperture (per exposure):

Source noise = 288.23

Background noise = 725.33

Dark current noise = 2.31

Readout noise = 41.22

Total noise per exposure = 781.59

Total signal per exposure = 83074.56

Intermediate S/N for one exposure = 106.29

S/N for the whole observation = 138.80 (including sky subtraction)

Requested total integration time = 120.00 secs, of which 120.00 secs is on source.

The peak pixel signal + background is 5933 e- (3295 ADU). This is 5% of the saturation limit of 117963 e-.

Observation Overheads

Setup	360.0 s	
Telescope offset	1 x 7.0 s	assuming ABAB dithering pattern
Exposure	2 x 60.0 s	
Readout	2 x 82.5 s	
DHS Write	2 x 10.0 s	
Program time	11 mins 12 secs	

Output:

- Spectra autoscaled.

Input Parameters:

Instrument: GMOS-S

Source spatial profile, brightness, and spectral distribution:

The z = 0.95000 point source is a 19.6 Vega user defined SED in the i band with the name: sn2011fe_exptime.dat

Instrument configuration:

Optical Components:

- Filter: z_G0328
- Fixed Optics
- Detector - Hamamatsu array

Amp gain: Low, Amp read mode: Slow

Region of Interest: Full Frame Readout

Binning: 1 × 1

Pixel Size in Spatial Direction: 0.080778arcsec

Telescope configuration:

- silver mirror coating.
- side looking port.

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Gemini Integration Time Calculator

- wavefront sensor: oiwfs

Observing Conditions:

- Airmass: 1.50
- Image Quality: 70% ($\leq 0.63''$ at zenith, $\leq 0.80''$ on-source, **exact condition specified**)
- Cloud cover: 70%
- Water Vapor: 100%
- Sky Background: 80%

Likelihood of execution: 39%

Calculation and analysis methods:

- Mode: imaging
- Calculation of S/N ratio with 2 exposures of 60.00 secs, and 100.00% of them on source.
- Analysis performed for aperture that gives 'optimum' S/N and a sky aperture that is 5.00 times the target aperture.